AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	30.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Burning urination;Frequency
Comorbidities: None
Surgical History: None
Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9000.0	cells/µL	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	9.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.1	mg/dL	3.5 - 7.2
Blood Urea	25.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	**Enterococcus faecium**
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	
Predicted Resistant	

Recommended Therapy

Clinical Summary

Patient: 30-year-old female, recurrent urethritis (lower urinary-tract infection). Presenting with dysuria and frequency; urinalysis shows pyuria, modest inflammation (CRP 9 mg/L, WBC 9 × 10■/L).

Predicted pathogen: *Enterococcus faecium* - Gram-positive cocci with a high likelihood of vancomycin resistance (VRE). No specific resistance or sensitivity data are available from the model.

Implications of resistance profile
• *E. faecium* frequently exhibits resistance to β-lactams, aminoglycosides, and clindamycin.
• Prior exposure to clindamycin and linezolid raises concern for emerging resistance, especially to linezolid.

Recommended empirical therapy (adjust after susceptibility results):

1. **Daptomycin** 6 mg/kg IV once daily (adjust for renal function).
• Active against VRE, bactericidal, rapid urinary excretion.
• Monitor CK weekly; avoid in patients with severe renal impairment without dose adjustment.

2. **If oral step-down is needed and susceptibility permits: Tedizolid** 200 mg PO once daily for 7-10 days.
• Similar spectrum to linezolid but lower myelosuppression risk.
• Watch for thrombocytopenia and drug-drug interactions (MAO inhibition).

3. **Alternative (if daptomycin unavailable and susceptibility confirmed): High-dose ampicillin** (2 g IV q4h) plus **gentamicin** (1-1.5 mg/kg IV q8h) *only* if the isolate is ampicillin-susceptible and synergy is demonstrated. Requires close renal monitoring and therapeutic drug monitoring for gentamicin.

Rationale: Daptomycin and tedizolid provide reliable coverage of VRE and avoid reliance on agents previously used (clindamycin, linezolid). Combination β-lactam/aminoglycoside is reserved for susceptible strains to minimize toxicity.

Key precautions & monitoring:
• Baseline and periodic CK (daptomycin) and CBC (tedizolid/linezolid).
• Renal function (creatinine, urine output) for aminoglycoside use.

• Assess for drug interactions (MAO inhibition with tedizolid/linezolid).
• Obtain definitive susceptibility testing from the pus swab; de-escalate therapy accordingly.

Actionable plan: Initiate IV daptomycin now, obtain culture and susceptibility results, and transition to oral tedizolid if sensitivities allow and the patient can tolerate oral therapy. Re-evaluate clinical response in 48-72 h.